

Evidence Summary - Technology #06

Neuromorphic Shunt Failure Predictor

External Evidence Sources (5 total):

Source 1: Implantable Intracranial Pressure Sensor with Continuous

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10532732>

Excerpt: by Y Elsawaf · 2023 · Cited by 4 — an ambulatory intracranial pressure (ICP) monitoring device may assist in the detection of shunt failure without an invasive diagnostic workup.

Source 2: Evolving concepts in intracranial pressure monitoring

URL: <https://www.sciencedirect.com/science/article/pii/S1878747924001946>

Excerpt: by R Mathur · 2025 · Cited by 20 — Intraparenchymal ICP monitors are an alternative to EVDs and use implantable micro-transducers to measure local ICP at the tip of the probe placed into the ... Read more

Source 3: Early fault prediction and detection of hydrocephalus

URL: https://file.scirp.org/Html/7-9101515_28918.htm

Excerpt: Long-term monitoring of intracranial pressure (ICP) is limited by the lack of an implantable sensor with low drift as well as the power needed for the implanted ... Read more

Source 4: Australia OKs coin-sized sensor for hydrocephalus

URL: <https://www.mobihealthnews.com/news/anz/australia-oks-coin-sized-sensor-hydrocephalus-management>

Excerpt: Mar 15, 2026 — The implant enables real-time intracranial pressure monitoring , cutting hospital scans by over 60%, monitoring procedures by 90%, and reducing ... Read more

Source 5: Emerging Technology

URL: <https://www.hydroassoc.org/emerging-technology>

Excerpt: The sensor responds to ultrasound signals and can be read using standard or handheld ultrasound devices. It is designed for long-term or acute ICP monitoring in ... Read more

*Evidence Classification: This dossier contains source-native information, derived engineering analysis, engineering proposals, and unresolved questions.
No evidence has been physically validated.*